FIT3181 2024 Sem 2 In Semester (Australia Malaysia) (10)

Financial Management (香港中文大學（深圳）)



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MONASH University

2024 Semester Two In Semester Examination Period

Faculty of Information Technology

EXAM CODES:	**FIT3181**
TITLE OF PAPER:	Deep learning
EXAM DURATION:	2 hours 10 mins

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Authorised Materials

CALCULATORS	☑ YES	☐ NO	Physical / Virtual Calculator
DICTIONARIES	☐ YES	☑ NO	
NOTES	☑ YES	☐ NO	Notes - Double sided A4 x 1 - Physical Only
WORKING SHEETS	☑ YES	☐ NO	
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if yes, items permitted are: Virtual Calculators: i) https://www.online-calculator.com/, ii) https://www.desmos.com/scientific, iii) https://www.calculator.net/scientific-calculator.html

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This examination is designed for the FIT3181 Deep Learning unit, S2 2024.

- This is a **closed book** exam with **specifically permitted items**.
- You are permitted to use an unlimited number of blank working sheets.
- You are permitted to bring: Notes - Double sided A4 x 1 - Physical Only
- A Physical Calculator and the following Virtual Calculators are permitted:
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- Please answer **ALL** questions.

It contains THREE (3) parts with a total of **100 marks**:

- **Part A** contains **12** multiple-choice questions, together they are worth a total of **20 marks**. For questions with more than one answer, a correct choice will receive a partial mark and incorrect choices will reduce the mark. To receive full marks, only correct choices must be selected.
- **Part B** contains **5** short workout and knowledge questions, worth **45 marks**. These questions typically require short knowledge answers and calculations based on the knowledge you have learned from the unit. Having the calculator handy for these questions is recommended.
- **Part C** contains **3** mixed and handwritten-answer questions, worth **35 marks**. These questions typically assess the knowledge and understanding of lecture contents.
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Part A - Multiple Choice Questions (12 questions, 20 marks)

Information

Part A consists of **12 multiple-choice questions**, worth 20 marks.

Question 1

1 Mark

Consider an image classification task with six classes {cat=1, dog=2, lion=3, flower=4, cow=5, car=6}. Consider an image x. Assume that a Convolutional Neural Network gives a prediction probabilities f(x)=[0.2, 0.15, 0.05, 0.1, 0.3, 0.2] and categorial ground-truth label of x is cow. What is the cross-entropy loss suffered by this prediction?

Select one:

- a. - log 0.2
- b. - log 0.1
- c. log 0.3
- d. log 0.1
- e. - log 0.3

Question 2

1 Mark

Assume that we have 5 classes in {cat = 1, dog = 2, lion = 3, monkey = 4, car = 5}. Given a data example x with ground-truth label dog, assume that a feed-forward NN gives discriminative scores to this x as h_1= -3, h_2=6, h_3= -2, h_4= -1, h_5=3. What is the probability to predict x as lion or p(y=lion | x)?

Select one:

- a. $\frac{e^{1}}{e^{-3}+e^{6}+e^{-2}+e^{-1}+e^{3}}$
- b. $\frac{e^{6}}{e^{-3}+e^{6}+e^{-2}+e^{-1}+e^{3}}$
- c. 1
- d. $\frac{e^{-1}}{e^{-3}+e^{6}+e^{-2}+e^{-1}+e^{3}}$
- e. $\frac{e^{-2}}{e^{-3}+e^{6}+e^{-2}+e^{-1}+e^{3}}$

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Question 3

1 Mark

Given an implementation of the ResNet as below. Assume that we are feeding a batch [16, 3, 64, 64] to our ResNet. What are the shape of A,B,C,D, E, and F?

```
class create_ResNet(nn.Module):
  def __init__(self):
    super().__init__()
    self.layers = nn.ModuleList([
                  nn.LazyConv2d(64, kernel_size=7, stride=2, padding=3),
                  nn.LazyBatchNorm2d(),
                  nn.ReLU(),
          A <---  nn.MaxPool2d(kernel_size=3, stride=2, padding=1),
          B <---  ResnetBlock(64, 2, first_block=True),
          C <---  ResnetBlock(128, 2),
          D <---  ResnetBlock(256, 2),
          E <---  nn.AdaptiveAvgPool2d((1, 1)),
                  nn.Flatten(1),
          F <---  nn.LazyLinear(10),
                  # nn.Softmax(dim=-1)
                  ])
  def forward(self, X):
    for _, layer in enumerate(self.layers):
      X = layer(X)
    return X
```

Select one:

- a. [16,64,16,16], [16,64,16,16], [16,128,8,8], [16,256,4,4], [16,256,1,1], [16,10]
- b. [16,64,16,16], [16,64,16,16], [16,128,8,8], [16,256,4,4], [16,256], [32,10]
- c. [16,64,16,16], [16,64,8,8], [16,128,4,4], [16,256,2,2], [16,256,1,1], [16,10]
- d. Raise an error.

Question 4

2 Marks

Given the loss function $J(w)=\frac{1}{1000}\sum_{i=1}^{1000}(w-x_i)^2$ where w is the model parameter and $x_i=i^2$ for i=1,….,1000 are the data points. We need to solve $\min_w J(w)$ using stochastic gradient descent with the learning rate η=0.2. Assume we sample a batch of indices $i_1=1$, $i_2=2$, $i_3=3$, $i_4=4$ and at the iteration t, we have $w_t=20$. What is the value of w_{t+1} at the next iteration?

Select one:

- a. 15
- b. 17
- c. 19
- d. 18
- e. 16

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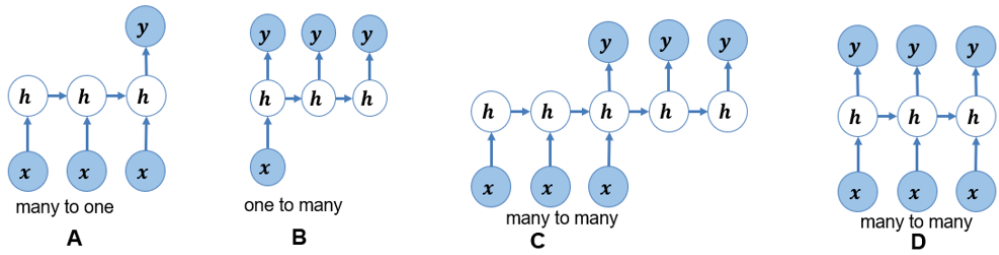
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Question 5

Matching RNN topologies and applications

1 Mark



Sentiment analysis	· C · D · A · B
Image captioning	· C · D · A · B
Machine translation	· C · D · A · B
Video classification at frame level	· C · D · A · B

Question 6

2 Marks

Given an 3D input tensor with shape [10, 64, 64] over which we apply a conv2D with 20 filters each of which has shape [10 , 5, 5], strides [2,2], and padding=1. What is the shape of the output tensor?

Select one:

- a. [20, 32, 32]
- b. [20, 31, 31]
- c. [31, 31, 10]
- d. [31, 31, 20]
- e. [32, 32, 20]

Question 7

2 Marks

Assume that the tensor before the last tensor of a CNN has shape [128, 10 , 32, 32] and we apply 8 filters each of which has the shape [10, 5, 5] and strides= [2,2] with padding = 0 to obtain the last tensor. What is the shape of the last tensor?

Select one:

- a. [128, 8, 14, 14]
- b. [128, 8, 15, 15]
- c. [128, 8, 16, 16]
- d. [128, 8, 17, 17]

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Question 8

2 Marks

Given a DL model f(x;θ) parameterized by θ where f(x;θ) represents the prediction probabilities of x associated with a ground-truth label y ∈ {1,…,M} , we find an adversarial example by

$$x_{adv} = argmax_{x' \in B_\epsilon(x)} l(f(x'; \theta), y)$$

Which statements are correct?

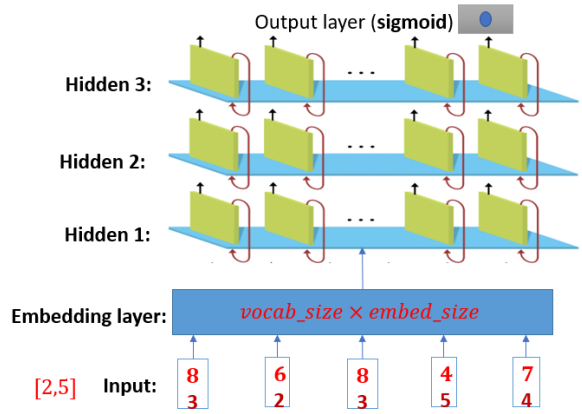
Select one or more:

- [ ] a. We maximally increase the chance to predict x with label y.
- [ ] b. It is a targeted attack.
- [ ] c. We maximally increase the chance to predict x with any else label y'≠y.
- [ ] d. We maximally decrease the chance to predict x with label y.
- [ ] e. It is an untargeted attack.

Question 9

2 Marks

Given an RNN with three hidden layers of GRU cells, assume that the sizes of hidden states on the hidden layers are 20, 30, and 40 respectively and sequence length of input and hidden layers is 5, and the embedding size of embedding layer is 10. We feed to our network a mini-batch of shape [2,5]. What are the shape of the output tensor of second hidden layer (i.e., hidden layer 2) if we consider all hidden states of this layer?



Select one:

- ◯ a. [2, 5, 10]
- ◯ b. [2, 5, 40]
- ◯ c. [2, 5, 20]
- ◯ d. [2, 5, 30]

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Question 10

2 Marks

Given a skip-gram model with vocabulary size 200 and embedding size 100, we consider a pair of target and context words with indices 8 and 10 respectively. Let U and V be two weight matrices connecting input to hidden layers and hidden to output layers. What statements are correct?

Select one or more:

- ☐ a. The hidden value h is the row 10 of the matrix U
- ☐ b. Input to the network is one-hot vector 1_{10}.
- ☐ c. Input to the network is one-hot vector 1_8
- ☐ d. The hidden value h is the row 8 of the matrix U
- ☐ e. Shape of U is [200,100] and shape of V is [100,200]
- ☐ f. Shape of U is [100,200] and shape of V is [200,100]

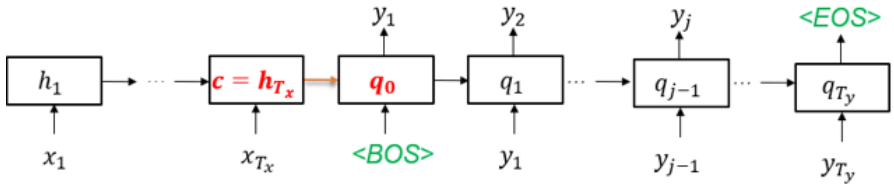
Question 11

2 Marks

In seq2seq for machine translation in the following figure, we derive the log-likelihood as follows:

$$P(\boldsymbol{y}|\boldsymbol{x},\theta) = P\left(\boldsymbol{y}_{1:T_y} \mid \boldsymbol{x}_{1:T_x},\theta\right) = P\left(\boldsymbol{y}_{1:T_y} \mid \boldsymbol{c},\theta\right) \overset{(1)}{\Rightarrow} P(\boldsymbol{y}_1 \mid \boldsymbol{c},\theta)P(\boldsymbol{y}_2 \mid \boldsymbol{y}_1,\boldsymbol{c},\theta)\ldots P(\boldsymbol{y}_j \mid \boldsymbol{y}_{1:j-1},\boldsymbol{c},\theta)\ldots P\left(\boldsymbol{y}_{T_y} \mid \boldsymbol{y}_{1:T_y-1},\boldsymbol{c},\theta\right)$$

$$= \prod_{i=1}^{T_y} P(\boldsymbol{y}_j \mid \boldsymbol{y}_{1:j-1},\boldsymbol{c},\theta) \overset{(2)}{\Rightarrow} \prod_{i=1}^{T_y} P(\boldsymbol{y}_j \mid \boldsymbol{q}_{j-1},\boldsymbol{c},\theta)$$



Which statements are correct?

Select one or more:

- ☐ a. In the derivation (2), q_{j-1} is viewed as a summary of the sequence $\boldsymbol{y}_{1:T_y}$
- ☐ b. In the derivation (1), c is viewed as a summary of the sequence $\boldsymbol{y}_{1:T_y}$
- ☐ c. In the derivation (1), c is viewed as a summary of the sequence $\boldsymbol{x}_{1:T_x}$
- ☐ d. $P(\boldsymbol{y}_j \mid \boldsymbol{q}_{j-1},\boldsymbol{c},\theta)$ means that on top of q_{j-1}, c, we can build up some dense layers to predict y_j
- ☐ e. In the derivation (2), q_{j-1} is viewed as a summary of the sequence $\boldsymbol{y}_{1:j-1}$

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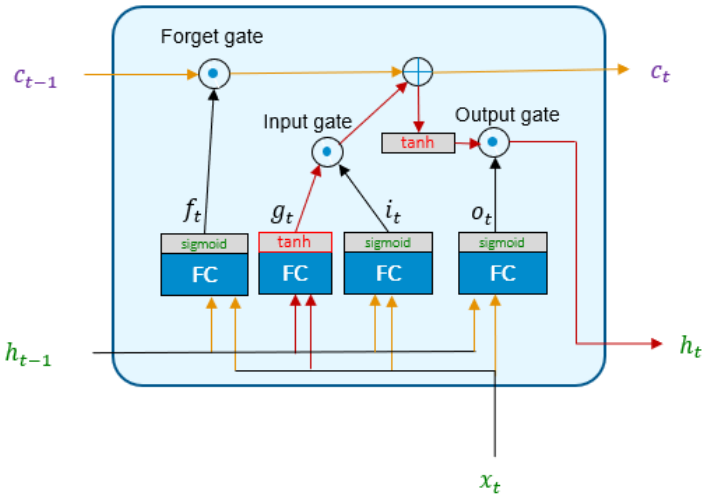
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Question 12

Consider LSTM cell which is shown in the following figure. What statements are correct for the forget gate?

2 Marks



Select one or more:

- [ ] a. It controls the proportion of information in long-term memory to be carried forward
- [ ] b. We apply sigmoid activation when computing forget gate
- [ ] c. We apply tanh activation when computing forget gate
- [ ] d. It controls the proportion of information in short-term memory to be carried forward

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Part B - Short Workout & Knowledge Questions (5 questions, 45 marks)

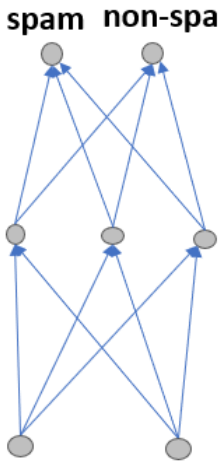
Information

Part B has 5 questions, worth 45 marks.

Question 13

10 Marks

Consider a feed-forward neural network as shown in the figure for the spam email detection with two labels (spam=1 and non-spam=2). Assume that we feed a feature vector x=[-1,1] with true label y=1 to the network.



$$h^2, p = softmax(h^2)$$

$$W^2 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}, b^2 = [-6 \ -7]$$

$$\bar{h}^1, h^1 = ReLU(\bar{h}^1)$$

$$W^1 = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix}, b^1 = [-2 \ 0 \ 1]$$

$$x = [-1,1], y = 1$$

a) What are the formulas and the values of $\bar{h}^1, h^1$? **[2 marks]**

b) What are the formulas and the values for the logit h^2 and the prediction probability p? **[2 marks]**

c) What is the predicted label $\hat{y}$ and the cross-entropy loss ℓ for this prediction? Is it a correct or incorrect prediction? **[2 marks]**

d) What is the derivative $\frac{\delta \ell}{\delta h^2}$? **[1 mark]**

e) What is the derivative $\frac{\delta \ell}{\delta W^2}$? **[1 mark]**

f) Assume that we are using Stochastic Gradient Descent with the learning rate η=0.1 to update W^2, what is the new weight matrix W^2? **[2 marks]**

Please answer the question on your blank piece of paper.

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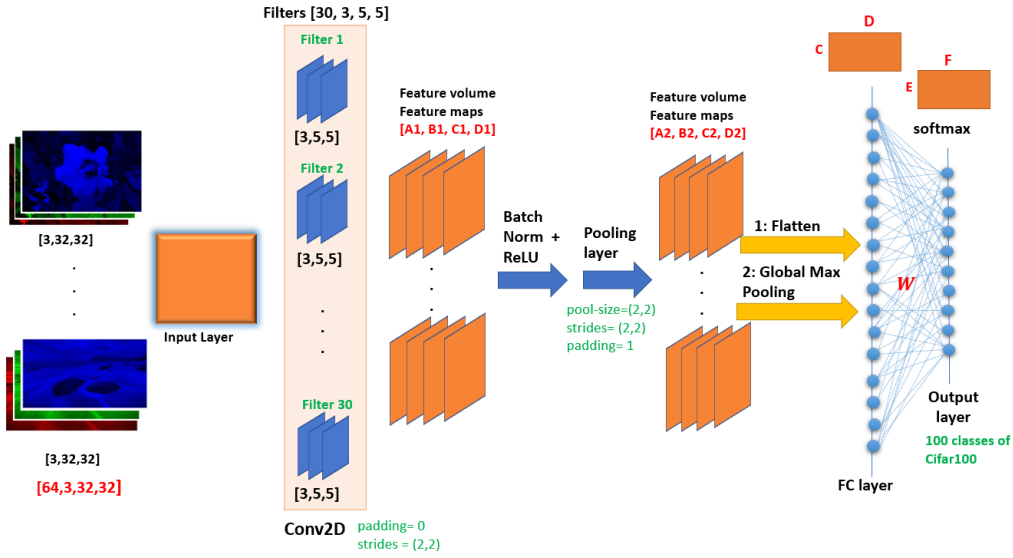
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Question 14

10 Marks

Assume that we conduct a Convolution Neural Network (CNN) with the configuration as shown in the below figure to predict the image dataset Cifar100 with 100 classes. We feed a batch of images with the shape **[64, 3, 32, 32]** to our CNN. Answer the following questions.



14a)

2 Marks

What is the shape of the feature maps [A1, B1, C1, D1]? Show the steps of your answer.

14b)

2 Marks

What is the shape of the feature maps [A2, B2, C2, D2]? Show the steps of your answer.

14c)

2 Marks

What is the shape of the 2D tensor [C,D] if we follow 1: Flatten? What does this 2D tensor store? Moreover, what is the shape of the weight matrix W in this case?

14d)

2 Marks

What is the shape of the 2D tensor [C,D] if we follow 2: Global Max Pooling? What does this 2D tensor store? Moreover, what is the shape of the weight matrix W in this case?

14e)

2 Marks

What is the shape of the 2D tensor [E,F]? What does this 2D tensor store?

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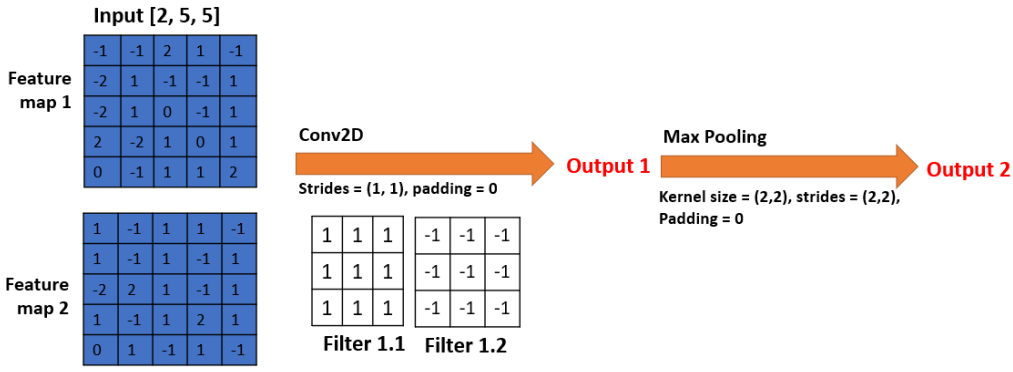
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Question 15

10 Marks

Assume that we have [2, 5, 5] input tensor as shown below. We first apply a Conv2D layer with one filter [2, 3, 3] (i.e., filter 1.1 and filter 1.2), strides = (1,1), and padding = 0 to obtain the output 1. We then apply max pooling with kernel size = (2,2), strides = (2,2), and padding= 0 to obtain the output 2.



15a)

5 Marks

What are the shape and the values of the feature maps in Output 1?

15b)

5 Marks

What are the shape and the values of the feature maps in Output 2?

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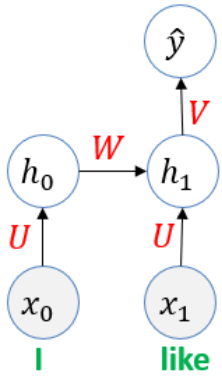
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Question 16

10 Marks

Assume that we have a RNN with two time-steps as in the figure. We feed to this RNN a sentence "**I like**" to predict a sentiment label $\hat{y}$ in the label set {1:positive, 2: negative}.



$$V=\begin{bmatrix}1 & 1\\ 0 & 1\\ 1 & 0\end{bmatrix}$$

$$U=\begin{bmatrix}1 & 1 & 0\\ 0 & 1 & 1\end{bmatrix} \text{ and } W=\begin{bmatrix}1 & 1 & 0\\ 0 & 1 & 1\\ 1 & 0 & 1\end{bmatrix}$$

Build up vocabulary

1. Like (index: 1)
2. Love (index: 2)
3. Bad (index: 3)
4. Fantastic (index: 4)
5. I (index: 5)
6. Recommend (index: 6)

Embedding matrix (E [6 × 2])

E_1	1	0.0
E_2	-1	1.3
E_3	1	3.3
E_4	-1	1.3
E_5	-1	1.0
E_6	-1.7	-1.3

a) What is the value of the hidden state h_0 if we do not use the bias and apply the ReLU activation function? **[3 marks]**

b) What is the value of the hidden state h_1 if we do not use the bias and apply the ReLU activation function? **[3 marks]**

c) On the top of h_1, we conduct a dense layer with the **weight matrix V** to predict the label $\hat{y}$. What are the prediction probabilities and the predicted label $\hat{y}$? **[4 marks]**

Please answer the question on your blank piece of paper.

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Question 17

Read the following code and provide the shape of the tensors h1, h2, h3, h4, h5, h6, h7, h8 if we input x [128, 3, 64, 64] to our CNN

5 Marks

```
class MyCNN(nn.Module):
    def __init__(self):
        super(MyCNN, self).__init__()
        self.conv1 = nn.Conv2d(3, 20, kernel_size= 3, stride= 2, padding=1)
        self.pool = nn.MaxPool2d(kernel_size=2, stride=2, padding =0)
        self.conv2 = nn.Conv2d(20, 32, kernel_size= 3, stride= 1, padding=1)
        self.conv3= nn.Conv2d(32, 64, kernel_size= 3, stride= 1, padding=1)
        self.avg_pool = nn.AdaptiveAvgPool2d((1, 1))
        self.flatten = nn.Flatten()
        self.fc1 = nn.Linear(64, 10)
    def forward(self, x):
        h1 = torch.relu(self.conv1(x))
        h2 = self.pool(h1)
        h3 = torch.relu(self.conv2(h2))
        h4 = self.pool(h3)
        h5 = torch.relu(self.conv3(h4))
        h6 = self.avg_pool(h5)
        h7 = self.flatten(h6)
        h8 = self.fc1(h7)
        return h8
```

```
myCNN = MyCNN()
x = torch.randn(128, 3, 64, 64)
y = myCNN(x)
```

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Part C - Mixed & Handwritten-Answer Questions (3 questions, 35 marks)

Information

Part C has 3 questions, worth 35 marks.

Question 18

5 Marks

Discuss THREE (3) differences and ONE (1) similarity between a convolutional layer (i.e., Conv2d) and a pooling layer (i.e., MaxPool2d or AvgPool2d).

Question 19

15 Marks

Consider a Convolution Neural Network (CNN) with the model parameter θ. Specifically, given an image x with the ground-truth label y, the CNN returns the prediction probabilities $f(x; \theta)$ over M classes and suffers the loss l ($f(x; \theta)$, y) where l is a loss function (e.g., the cross-entropy loss).

a) Adversarial examples are a serious issue for CNNs. Give a definition of adversarial examples. Give a practical example to explain why adversarial examples circumvent the applications of CNNs in reality. **[3 marks]**

b) Given a benign example x and the ϵ-ball B_ϵ = {x': $||x' - x||_\infty \leq \epsilon$}, describe and give the formula to find out a targeted adversarial example for x. **[3 marks]**

c) Given a benign example x and the ϵ-ball B_ϵ = {x': $||x' - x||_\infty \leq \epsilon$}, describe and give the formula to find out a untargeted adversarial example for x. **[3 marks]**

d) Given a mini-batch B = { (x_1, y_1), … , (x_b, y_b)} at an iteration, describe to perform adversarial training for this mini-batch to improve model robustness. **[3 marks]**

e) How adversarial training is similar to data augmentation? How adversarial training is different from data augmentation? **[3 marks]**

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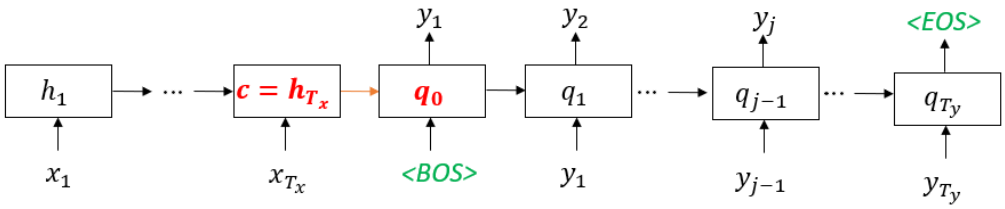
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Question 20

15 Marks

Consider a **seq2seq model** using the encoder-decoder architecture shown in the figure below. Let θ_e and θ_d be the parameters for the encoder and the decoder, respectively. Answer the following questions:



a) Explain the role of the **BOS** and **EOS** symbols? **[2 marks]**

b) Let $\theta = [\theta_e, \theta_d]$ be the parameter vector to be learned and D be the training set consists of multiple pairs of sequences **(x, y)**, write down the maximum log-likelihood objective function to learn θ. **[2 marks]**

c) Use the product rule, expand the log-likelihood term **log P(y | x,θ)** where $y=[y_1,\ldots,y_{Ty}]$ and $x=[x_1,\ldots,x_{Tx}]$ so that it can be computed based on the **local probability** $P(y_j \mid q_{j-1}, c, \theta)$. What is the meaning of the **local probability** $P(y_j \mid q_{j-1}, c, \theta)$? How can we realize this local probability in an implementation? **[3 marks]**

d) During inference, given an **input sequence x** , what are TWO (2) common strategies to infer corresponding output sequence y? According to you, which strategy is better and explain why? **[3 marks]**

e) What is the main drawback of this time-invariant context vector c? Present the global attention mechanism and how this can help improve seq2seq models. **[5 marks]**

Please answer the question on your blank piece of paper.

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